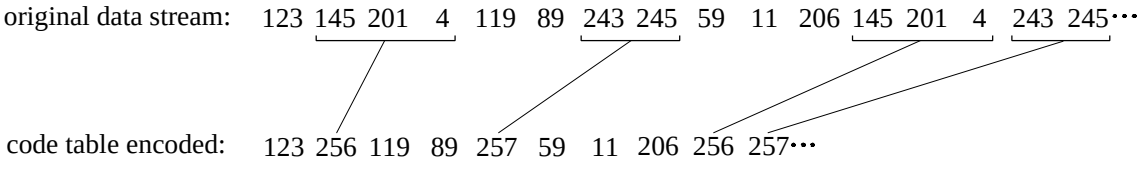
Example Code Table

| | code number | translation |
|---|---|---|
| *identical code* | 0000 | 0 |
| | 0001 | 1 |
| | ⋮ | ⋮ |
| | 0254 | 254 |
| | 0255 | 255 |
| *unique code* | 0256 | 145 201 4 |
| | 0257 | 243 245 |
| | ⋮ | ⋮ |
| | 4095 | xxx xxx xxx |

FIGURE 27-6
Example of code table compression. This is the basis of the popular LZW compression method. Encoding occurs by identifying sequences of bytes in the original file that exist in the code table. The 12 bit code representing the sequence is placed in the compressed file instead of the sequence. The first 256 entries in the table correspond to the single byte values, 0 to 255, while the remaining entries correspond to *sequences* of bytes. The LZW algorithm is an efficient way of generating the code table based on the particular data being compressed. (The code table in this figure is a simplified example, not one actually generated by the LZW algorithm).




The LZW method achieves compression by using codes 256 through 4095 to represent *sequences* of bytes. For example, code 523 may represent the sequence of three bytes: 231 124 234. Each time the compression algorithm encounters this sequence in the input file, code 523 is placed in the encoded file. During uncompression, code 523 is translated via the code table to recreate the true 3 byte sequence. The longer the sequence assigned to a single code, and the more often the sequence is repeated, the higher the compression achieved.

Although this is a simple approach, there are two major obstacles that need to be overcome: (1) how to determine what sequences should be in the code table, and (2) how to provide the uncompression program the same code table used by the compression program. The LZW algorithm exquisitely solves both these problems.

When the LZW program starts to encode a file, the code table contains only the first 256 entries, with the remainder of the table being blank. This means that the first codes going into the compressed file are simply the single bytes from the input file being converted to 12 bits. As the encoding continues, the LZW algorithm identifies repeated sequences in the data, and adds them to the code table. Compression starts the second time a sequence is encountered. The key point is that a sequence from the input file is not added to the code table until it has already been placed in the compressed file as individual characters (codes 0 to 255). This is important because it allows the uncompression program to *reconstruct* the code table directly from the compressed data, without having to transmit the code table separately.